

Qualification Pack



Dredge and Ore Processing Operator

QP Code: MIN/Q1803

Version: 1.0

NSQF Level: 4.0

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MIN/Q1803: Dredge and Ore Processing Operator

Brief Job Description

A Dredge and Ore Processing Operator is responsible for extracting ores from dredge, sending the slurry to wet concentrator unit, supplementary mining plant and spiral operation, tailing management and dumping.

Personal Attributes

This job requires the individual to concentrate on the job at hand and complete it efficiently and effectively. The individual should be physically agile, strong, have good eye-sight and not suffer from colorblindness.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [MIN/N4107: Carry out the dredging operation](#)
2. [MIN/N4108: Carry out the upgradation unit operations](#)
3. [MIN/N1812: Carry out repair and maintenance of Heavy Earth Moving Machinery \(HEMM\)](#)
4. [MIN/N1813: Perform inventory and logistics management](#)
5. [MIN/N1810: Follow the applicable health, safety and environmental guidelines \(Placer Mining\)](#)
6. [MIN/N1811: Work collaboratively and effectively in a team](#)
7. [DGT/VSQ/N0102: Employability Skills \(60 Hours\)](#)

Qualification Pack (QP) Parameters

Sector	Mining
Sub-Sector	Mining Operation
Occupation	Placer Mining
Country	India
NSQF Level	4.0
Credits	15

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Aligned to NCO/ISCO/ISIC Code	NCO-2015/ 8342.0900, NCO-2015/ 3121.9900
Minimum Educational Qualification & Experience	12th grade Pass OR Completed 2nd year of the 3-year diploma after 10 (in Mining/ Chemical/ Metallurgy/ Material Science) OR 10th grade pass (with two years of NTC/ NAC/ CITS or equivalent (in Mechanical/ Electrical)) OR 11th grade pass (with 1 year relevant experience) OR 10th grade pass (with 2 years relevant experience) OR 8th grade pass with 2 year NTC plus 1 year NAC plus 1 year CITS (in Mechanical/ Electrical) OR Previous relevant Qualification of NSQF Level (3.0)(with 3 years relevant experience) OR Previous relevant Qualification of NSQF Level (3.5)(with 1.5 year relevant experience)
Minimum Level of Education for Training in School	
Pre-Requisite License or Training	NA
Minimum Job Entry Age	18 Years
Last Reviewed On	NA
Next Review Date	14/03/2027
NSQC Approval Date	15/03/2024
Version	1.0
Reference code on NQR	QG-04-MG-02172-2024-V1- SCMS
NQR Version	1.0

Remarks:

450 Hours

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MIN/N4107: Carry out the dredging operation

Description

This occupational standards unit is about carrying out the dredging operation

Scope

The scope covers the following :

- Check the dredge operation

Elements and Performance Criteria

Check the dredge operation

To be competent, the user/individual on the job must be able to:

- PC1.** select the appropriate type of dredge for use in the plant
- PC2.** ensure the dredger unit is in complete floating condition and there is no grounding of the Unit at any point
- PC3.** check the dredger unit is appropriately positioned and anchored at all corners using winch ropes and spuds
- PC4.** transport the materials in the maintenance barge
- PC5.** assess the weight of the material and keep the load in the designated place with appropriate anchorage
- PC6.** check the slurry pump for the correct functioning
- PC7.** ensure the dredger pump is appropriately primed and the suction pipe is completely submerged inside water before starting the dredger pump
- PC8.** check and ensure all the operating parameters of the dredger unit, e.g. main motor temperature, mass flow meter reading, main motor ammeter reading, head feed bin level indicator reading, etc., are within the permissible limits
- PC9.** maintain the appropriate flow rate and pulp density of the dredger to match with the requirements for the operation of the up-gradation unit
- PC10.** maximize the dredge cutting depth to six meters below the surface of the water over the entire area by monitoring the depth indicator and by swinging mechanism to get the uniform pond bottom profile for the ease of plant shifting
- PC11.** maintain the required suction vacuum and discharge pressure of the dredger pump by using the pressure indicators provided on the unit
- PC12.** convey the slurry to the wet up-gradation unit maintaining parameters, such as feed rate, pulp density, head feed bin indicator, mass flow meter reading, etc
- PC13.** check the layout and condition of the components of the hydraulic system, such as the hydraulic power pack, accumulator, actuators, check valve, pressure control valve, flow control and direction control valves, hoses and their coupling
- PC14.** conduct operational checks on the areas of potential issues, e.g. unusual noises/ vibrations, blockages and obstruction, lubrication of drive head
- PC15.** ensure the surge bin level is within the safe operating range

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PC16. flush the pumps with clean water following the standard procedure

PC17. follow the appropriate procedure for stopping the dredge pump, i.e. first stop the dredge pump and then hydraulic and other pumps

Follow equipment safety measures

To be competent, the user/individual on the job must be able to:

PC18. use the appropriate safety gear for anchor block shifting, winch wire rope handling, locking of winches, sandblasting, etc

PC19. maintain a safe distance from the pond edge in front of the dredge and avoid walking on the pontoons during the movement of dredge

PC20. follow the Safety Management Plan (SMP) and Emergency Management Plan (EMP) and encourage others to do the same

PC21. ensure the dredge deck floor is clean and tidy by washing away the spread oil, grease etc., using a water jet

PC22. follow the appropriate safety measures while working at haul roads, heights, dumps, sump area, stockyards, near moving parts, etc

PC23. ensure no unauthorized personnel is present on the maintenance barge during material handling

PC24. follow the standard signal while lifting materials

PC25. ensure coupling guards/ belt guards of pumps are in the correct position

PC26. ensure to switch off the lighting distribution board while cleaning the spiral with a water jet

PC27. use appropriate chain pulley, slings and ropes for loading and unloading materials

PC28. follow appropriate measures to avoid coming in contact with noxious fumes/ gases

PC29. maintain constant contact with the man standing outside through VHF wireless while dewatering inside the deck compartment

PC30. use headlamp of the recommended specification with proper gauge protection during deck cleaning

PC31. maintain the pontoon hand rails in good condition

PC32. use and follow caution and safety signs, stickers, and tags

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. beach sand deposit and mineral sand mining

KU2. the difference between conventional mining and mineral sand mining

KU3. different types of mining methods

KU4. the working principle of slurry pump

KU5. the importance of following the daily maintenance checklist in dredging operation

KU6. mine organization and the importance of discipline and punctuality in mine operations

KU7. the importance of using the mining equipment appropriately and loss to the company due to equipment damage or breakdown

KU8. the code of practice applicable to different areas of mining

KU9. the applicable safety regulations and procedures, and the impact of violating them

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- KU10.** the relevant process parameters, such as voltage, oil temperature, pump pressure and their operational value
- KU11.** different types of mining methods and dredge
- KU12.** the importance of health and safety in the mining operations
- KU13.** the relevant hazards and safety aspects in dredging activities
- KU14.** the handling reagents and using the relevant PPE
- KU15.** the appropriate emergency response/disaster management plan for accidents in mining operations

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** carry out appropriate documentation concerning work
- GS2.** read equipment manuals and process documents to understand the equipment and processes
- GS3.** listen attentively to understand the instructions or information being shared
- GS4.** communicate politely and professionally
- GS5.** plan and carry out different activities appropriately as per the applicable timelines and quality norms
- GS6.** follow the task lists and schedules
- GS7.** make appropriate decisions concerning the areas of work, following the organizational authority matrix
- GS8.** follow the given instructions and work on the identified areas of improvement
- GS9.** evaluate the possible solutions to work-related problems by applying reasoning skills and select the best one
- GS10.** follow the organizational procedure during emergencies

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Check the dredge operation</i>	16	30	-	11
PC1. select the appropriate type of dredge for use in the plant	1	2	-	1
PC2. ensure the dredger unit is in complete floating condition and there is no grounding of the Unit at any point	1	1	-	-
PC3. check the dredger unit is appropriately positioned and anchored at all corners using winch ropes and spuds	1	3	-	1
PC4. transport the materials in the maintenance barge	1	1	-	-
PC5. assess the weight of the material and keep the load in the designated place with appropriate anchorage	-	1	-	1
PC6. check the slurry pump for the correct functioning	1	3	-	1
PC7. ensure the dredger pump is appropriately primed and the suction pipe is completely submerged inside water before starting the dredger pump	1	2	-	-
PC8. check and ensure all the operating parameters of the dredger unit, e.g. main motor temperature, mass flow meter reading, main motor ammeter reading, head feed bin level indicator reading, etc., are within the permissible limits	1	3	-	1
PC9. maintain the appropriate flow rate and pulp density of the dredger to match with the requirements for the operation of the up-gradation unit	1	1	-	1
PC10. maximize the dredge cutting depth to six meters below the surface of the water over the entire area by monitoring the depth indicator and by swinging mechanism to get the uniform pond bottom profile for the ease of plant shifting	1	1	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC11. maintain the required suction vacuum and discharge pressure of the dredger pump by using the pressure indicators provided on the unit	1	1	-	1
PC12. convey the slurry to the wet up-gradation unit maintaining parameters, such as feed rate, pulp density, head feed bin indicator, mass flow meter reading, etc	1	1	-	1
PC13. check the layout and condition of the components of the hydraulic system, such as the hydraulic power pack, accumulator, actuators, check valve, pressure control valve, flow control and direction control valves, hoses and their coupling	1	3	-	1
PC14. conduct operational checks on the areas of potential issues, e.g. unusual noises/ vibrations, blockages and obstruction, lubrication of drive head	1	3	-	1
PC15. ensure the surge bin level is within the safe operating range	1	1	-	-
PC16. flush the pumps with clean water following the standard procedure	1	1	-	-
PC17. follow the appropriate procedure for stopping the dredge pump, i.e. first stop the dredge pump and then hydraulic and other pumps	1	2	-	1
<i>Follow equipment safety measures</i>	14	20	-	9
PC18. use the appropriate safety gear for anchor block shifting, winch wire rope handling, locking of winches, sandblasting, etc	1	1	-	-
PC19. maintain a safe distance from the pond edge in front of the dredge and avoid walking on the pontoons during the movement of dredge	1	2	-	1
PC20. follow the Safety Management Plan (SMP) and Emergency Management Plan (EMP) and encourage others to do the same	1	1	-	1
PC21. ensure the dredge deck floor is clean and tidy by washing away the spread oil, grease etc., using a water jet	1	2	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC22. follow the appropriate safety measures while working at haul roads, heights, dumps, sump area, stockyards, near moving parts, etc	1	1	-	1
PC23. ensure no unauthorized personnel is present on the maintenance barge during material handling	1	2	-	1
PC24. follow the standard signal while lifting materials	-	1	-	1
PC25. ensure coupling guards/ belt guards of pumps are in the correct position	1	2	-	-
PC26. ensure to switch off the lighting distribution board while cleaning the spiral with a water jet	1	2	-	1
PC27. use appropriate chain pulley, slings and ropes for loading and unloading materials	1	1	-	-
PC28. follow appropriate measures to avoid coming in contact with noxious fumes/ gases	1	1	-	1
PC29. maintain constant contact with the man standing outside through VHF wireless while dewatering inside the deck compartment	1	1	-	1
PC30. use headlamp of the recommended specification with proper gauge protection during deck cleaning	1	1	-	1
PC31. maintain the pontoon hand rails in good condition	1	1	-	-
PC32. use and follow caution and safety signs, stickers, and tags	1	1	-	-
NOS Total	30	50	-	20

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National Occupational Standards (NOS) Parameters

NOS Code	MIN/N4107
NOS Name	Carry out the dredging operation
Sector	Mining
Sub-Sector	Mineral Beneficiation
Occupation	Ore Processing
NSQF Level	4
Credits	3
Version	1.0
Last Reviewed Date	15/03/2024
Next Review Date	14/03/2027
NSQC Clearance Date	15/03/2024

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MIN/N4108: Carry out the upgradation unit operations

Description

This occupational standards unit is about the upgradation unit operations.

Scope

The scope covers the following :

- Operate the trommel
- Operate the static and vibrating screens
- Perform the preventive maintenance of vibrating screen
- Perform spiral separation
- Transfer concentrate minerals to MSP
- Manage the tailing

Elements and Performance Criteria

Operate the trommel

To be competent, the user/individual on the job must be able to:

- PC1.** check the layout and components of the trommel System, including drive mechanism, rollers, guards, screen, trommel tyres, feed end and discharge chute, roller bearings, controls such as fusible plug-in fluid coupling, etc
- PC2.** check the operations of mechanical and electrical driving units including safety devices
- PC3.** ensure the roller bearings, feed/discharge ends, drive mechanisms, lubrication systems, and controls drive belts, etc., of the trommel are appropriately maintained
- PC4.** feed sand slurry maintaining the appropriate parameters, such as feed-rate, ampere meter reading of drive motor, suction and discharge pressure of main pump, head feed bin level indicator, mass flow meter reading, etc.
- PC5.** conduct the screening of foreign material
- PC6.** convey the sand slurry through the trommel underflow chute to the surge bin below the trommel, and thereafter pump the same to the spirals
- PC7.** maintain appropriate parameters, e.g. feed/ material rate, alignment and inclination of the trommel, RPM, screen jamming, etc., while conveying sand slurry
- PC8.** maintain the cleanliness of the screens in the trommel and apertures to prevent blockages/jamming and maintain the required surge in the surge bin for the spiral operation
- PC9.** follow the safe code of practice for the erection, installation, operation, repairs, maintenance, and dismantling of the screens and the drive mechanism to prevent accidents
- PC10.** follow appropriate safety measures concerning the trommel operations

Operate the static and vibrating screens

To be competent, the user/individual on the job must be able to:

- PC11.** check the process of screening in supplementary mining plant
- PC12.** check the layout of components and power supply unit of vibrating screen, drive mechanism, bearing, feed arrangements, etc

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PC13. follow the manufacturer's instructions for the safe handling and maintenance of equipment

PC14. identify and remove the blockages in slurry and booster pumps

Perform the preventive maintenance of vibrating screen

To be competent, the user/individual on the job must be able to:

PC15. carry out greasing and lubrication of the moving parts of the equipment

PC16. follow the maintenance schedule recommended by equipment manufacturer

Perform spiral separation

To be competent, the user/individual on the job must be able to:

PC17. check the working of spirals, conducting checks on various operational parameters, e.g. monitors pulp density, feed rate, splitter position, diverter position, etc

PC18. ensure pump feed to the spiral at the required rate as per the process requirement

PC19. conduct separation to extract heavy minerals and lighter minerals

PC20. check for appropriate flow of materials through various spiral circuits, such as roughing, scavenging, cleaning, re-cleaning and control the speed and flow of material by varying the RPM of pumps

PC21. identify hazards, such as spills and leaks and take appropriate corrective measures, and report the condition to the supervisor

PC22. use safety signs at the locations of spills and leaks

Transfer concentrate minerals to MSP

To be competent, the user/individual on the job must be able to:

PC23. ensure that the heavies or the concentrator produced in the wet upgradation plant is conveyed through a series of booster pumps and pipelines to MSP

PC24. conduct operational checks for potential issues, such as suction/ discharge pressure, noise, vibrations, bearing heating, belt alignment, belt and tension of pumps as per the SOP

Manage the tailing

To be competent, the user/individual on the job must be able to:

PC25. direct the tailing to the backside of the upgradation plant in the form of slurry with proper densification, after the separation of the heavy minerals

PC26. ensure suitable conditions for single/distance tails jetting and its adoption

PC27. ensure the safety of the tailing area

PC28. ensure that no EME operator is working in the tailing dumps before starting the tailing pumps

PC29. monitor for the unintentional entry of person in the tailing area

PC30. use the Dozer for levelling and bring it to the near-original shape after complete stacking of the tailing heap

PC31. adopt plantation in the mined-out area

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. the importance and process of checking the layout and components of the trommel system, including drive mechanism, rollers, guards, screen, trommel tyres, feed end and discharge chute, roller bearings, controls such as fusible plug-in fluid coupling, etc

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- KU2.** how to check the operations of mechanical and electrical driving units including safety devices
- KU3.** the importance of maintaining the roller bearings, feed/discharge ends, drive mechanisms, lubrication systems, and controls drive belts, etc
- KU4.** the importance of maintaining the appropriate parameters, such as feed-rate, ampere meter reading of drive motor, suction and discharge pressure of main pump, head feed bin level indicator, mass flow meter reading, etc., while feeding sand slurry
- KU5.** the process of screening the foreign material
- KU6.** the process of conveying the sand slurry through the trommel underflow chute to the surge bin located below the trommel, and thereafter pumping the same to the spirals
- KU7.** the appropriate parameters to be maintained while conveying slurry
- KU8.** the importance of maintaining the cleanliness of the screens in the trommel and apertures
- KU9.** how to maintain the appropriate surge in the surge bin for the spiral operation
- KU10.** the safety practices to be followed for the erection, installation, operation, repairs, maintenance, and dismantling of the screens and the drive mechanism
- KU11.** the process of screening in the supplementary mining plant
- KU12.** the process of checking the layout of components and power supply unit of vibrating screen, drive mechanism, bearing, feed arrangements, etc
- KU13.** how to remove the blockages in slurry and booster pumps
- KU14.** the greasing and lubrication of the moving parts of the equipment
- KU15.** the importance of following the maintenance schedule recommended by equipment manufacturer
- KU16.** the importance and process of checking the working of spirals, conducting checks on various operational parameters, e.g. monitors pulp density, feed rate, splitter position, diverter position, etc
- KU17.** the process of conducting separation to extract heavy and lighter minerals
- KU18.** the importance of appropriate flow of materials through various spiral circuits, such as roughing, scavenging, cleaning, re-cleaning
- KU19.** the process of controlling the speed and flow of material by varying the RPM of pumps
- KU20.** advantage and disadvantage of tail densification system
- KU21.** the conditions suitable for single/distance tails jetting and its adoption

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** carry out appropriate documentation concerning work
- GS2.** read equipment manuals and process documents to understand the equipment and processes
- GS3.** listen attentively to understand the instructions or information being shared
- GS4.** communicate politely and professionally
- GS5.** plan and carry out different activities appropriately as per the applicable timelines and quality norms
- GS6.** follow the task lists and schedules



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- GS7.** make appropriate decisions concerning the areas of work, following the organizational authority matrix
- GS8.** follow the given instructions and work on the identified areas of improvement
- GS9.** evaluate the possible solutions to work-related problems applying reasoning skills and select the best one
- GS10.** follow the organizational procedure during emergencies

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Operate the trommel</i>	10	17	-	6
PC1. check the layout and components of the trommel System, including drive mechanism, rollers, guards, screen, trommel tyres, feed end and discharge chute, roller bearings, controls such as fusible plug-in fluid coupling, etc	1	3	-	1
PC2. check the operations of mechanical and electrical driving units including safety devices	1	3	-	1
PC3. ensure the roller bearings, feed/discharge ends, drive mechanisms, lubrication systems, and controls drive belts, etc., of the trommel are appropriately maintained	1	1	-	-
PC4. feed sand slurry maintaining the appropriate parameters, such as feed-rate, ampere meter reading of drive motor, suction and discharge pressure of main pump, head feed bin level indicator, mass flow meter reading, etc.	1	1	-	-
PC5. conduct the screening of foreign material	1	3	-	1
PC6. convey the sand slurry through the trommel underflow chute to the surge bin below the trommel, and thereafter pump the same to the spirals	1	2	-	1
PC7. maintain appropriate parameters, e.g. feed/material rate, alignment and inclination of the trommel, RPM, screen jamming, etc., while conveying sand slurry	1	1	-	1
PC8. maintain the cleanliness of the screens in the trommel and apertures to prevent blockages/jamming and maintain the required surge in the surge bin for the spiral operation	1	1	-	1
PC9. follow the safe code of practice for the erection, installation, operation, repairs, maintenance, and dismantling of the screens and the drive mechanism to prevent accidents	1	1	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. follow appropriate safety measures concerning the trommel operations	1	1	-	-
<i>Operate the static and vibrating screens</i>	4	9	-	3
PC11. check the process of screening in supplementary mining plant	1	3	-	1
PC12. check the layout of components and power supply unit of vibrating screen, drive mechanism, bearing, feed arrangements, etc	1	3	-	1
PC13. follow the manufacturer's instructions for the safe handling and maintenance of equipment	1	1	-	-
PC14. identify and remove the blockages in slurry and booster pumps	1	2	-	1
<i>Perform the preventive maintenance of vibrating screen</i>	2	2	-	1
PC15. carry out greasing and lubrication of the moving parts of the equipment	1	1	-	1
PC16. follow the maintenance schedule recommended by equipment manufacturer	1	1	-	-
<i>Perform spiral separation</i>	6	8	-	5
PC17. check the working of spirals, conducting checks on various operational parameters, e.g. monitors pulp density, feed rate, splitter position, diverter position, etc	1	2	-	1
PC18. ensure pump feed to the spiral at the required rate as per the process requirement	1	1	-	-
PC19. conduct separation to extract heavy minerals and lighter minerals	1	1	-	1
PC20. check for appropriate flow of materials through various spiral circuits, such as roughing, scavenging, cleaning, re-cleaning and control the speed and flow of material by varying the RPM of pumps	1	2	-	1

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC21. identify hazards, such as spills and leaks and take appropriate corrective measures, and report the condition to the supervisor	1	1	-	1
PC22. use safety signs at the locations of spills and leaks	1	1	-	1
<i>Transfer concentrate minerals to MSP</i>	2	4	-	2
PC23. ensure that the heavies or the concentrator produced in the wet upgradation plant is conveyed through a series of booster pumps and pipelines to MSP	1	2	-	1
PC24. conduct operational checks for potential issues, such as suction/ discharge pressure, noise, vibrations, bearing heating, belt alignment, belt and tension of pumps as per the SOP	1	2	-	1
<i>Manage the tailing</i>	6	10	-	3
PC25. direct the tailing to the backside of the upgradation plant in the form of slurry with proper densification, after the separation of the heavy minerals	1	2	-	1
PC26. ensure suitable conditions for single/distance tails jetting and its adoption	1	1	-	-
PC27. ensure the safety of the tailing area	-	2	-	1
PC28. ensure that no EME operator is working in the tailing dumps before starting the tailing pumps	1	2	-	1
PC29. monitor for the unintentional entry of person in the tailing area	1	1	-	-
PC30. use the Dozer for levelling and bring it to the near-original shape after complete stacking of the tailing heap	1	1	-	-
PC31. adopt plantation in the mined-out area	1	1	-	-
NOS Total	30	50	-	20

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National Occupational Standards (NOS) Parameters

NOS Code	MIN/N4108
NOS Name	Carry out the upgradation unit operations
Sector	Mining
Sub-Sector	Mineral Beneficiation
Occupation	Ore Processing
NSQF Level	4.0
Credits	3
Version	1.0
Last Reviewed Date	15/03/2024
Next Review Date	14/03/2027
NSQC Clearance Date	15/03/2024

Qualification Pack

MIN/N1812: Carry out repair and maintenance of Heavy Earth Moving Machinery (HEMM)

Description

This unit is about performing the repair and maintenance of HEMM for its optimal performance.

Scope

The scope covers the following :

- Maintain the HEMM

Elements and Performance Criteria

Maintain the HEMM

To be competent, the user/individual on the job must be able to:

- PC1.** check different types of HEMM used in OSCOM mining areas, such as dozer, loader, excavator etc. for land preparation, anchor block shifting, tails dozing and levelling material handling, skid-based structural shifting
- PC2.** conduct scheduled and routine examination of the vehicle to identify damage, corrosion, inadequate fluid levels, leaks, and wear
- PC3.** review the equipment's complaint sheet, log book and history sheet to determine the repair requirements
- PC4.** use the onboard digital diagnostic tools to assess the problem
- PC5.** use the equipment data obtained from pressure gauges, filler gauges, callipers, condition monitoring devices and compare it with the standards to detect faults in the system
- PC6.** check the lubricant levels and refill as required, ensuring to use the lubricant of the correct code and grade
- PC7.** follow the Original Equipment Manufacturer (OEM)-recommended procedure and checklist for routine servicing
- PC8.** calibrate, align and adjust the settings, alignment, pressures, tension, speeds and levels of the engine, transmission system, load bearing arms and structure, electrical and electronic components, valves, ignition, fuel and emissions, brakes, lights, tires, steering and body fittings
- PC9.** identify wear and tear in components, such as oil and air filters, drive belts, braking system due to continuous usage and replace them with new ones
- PC10.** follow the appropriate measures to avoid damage to the vehicle and its components during repair and maintenance
- PC11.** carry out breakdown maintenance at the worksite or workshop, as required
- PC12.** follow appropriate safety measures while operating and maintaining the HEMM, and hydraulic machine

Knowledge and Understanding (KU)

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The individual on the job needs to know and understand:

- KU1.** the importance of following the daily maintenance checklist for an HEMM operator
- KU2.** the impact of not following defined procedures/work instructions
- KU3.** the importance of using the HEMM appropriately and loss to the company due to equipment damage or breakdown
- KU4.** the safety guidelines specified by Directorate General of Mines Safety (DGMS) for HEMM operations
- KU5.** the code of practice in specific areas of the mine
- KU6.** different types of HEMM and their importance
- KU7.** the importance of HEMM being operated by the trained and authorized personnel
- KU8.** the appropriate speed for operating HEMM, i.e. up to 20 km/hour
- KU9.** the importance of checking seat belt, rear view mirror, braking system, oil gauge, temperature gauge, horn, lighting system, fire extinguisher etc. before operating HEMM
- KU10.** the process of identifying and rectifying faults in the HEMM with the help of a mechanical engineer
- KU11.** the appropriate health and safety measures for operating HEMM
- KU12.** the safe procedure of climbing a ladder, i.e. use of hand holds and railings
- KU13.** the safe method of overtaking other vehicles while operating a HEMM operator
- KU14.** the appropriate PPE to be used during the maintenance of the HEMM

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** carry out appropriate documentation concerning work
- GS2.** read equipment manuals and process documents to understand the equipment and processes
- GS3.** listen attentively to understand the instructions or information being shared
- GS4.** communicate politely and professionally
- GS5.** plan and carry out different activities appropriately as per the applicable timelines and quality norms
- GS6.** follow the task lists and schedules
- GS7.** make appropriate decisions concerning the areas of work, following the organizational authority matrix
- GS8.** follow the given instructions and work on the identified areas of improvement
- GS9.** evaluate the possible solutions to work-related problems applying reasoning skills and select the best one
- GS10.** follow the organizational procedure during emergencies

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Maintain the HEMM</i>	30	50	-	20
PC1. check different types of HEMM used in OSCOM mining areas, such as dozer, loader, excavator etc. for land preparation, anchor block shifting, tails dozing and levelling material handling, skid-based structural shifting	3	5	-	2
PC2. conduct scheduled and routine examination of the vehicle to identify damage, corrosion, inadequate fluid levels, leaks, and wear	3	5	-	2
PC3. review the equipment's complaint sheet, log book and history sheet to determine the repair requirements	3	4	-	2
PC4. use the onboard digital diagnostic tools to assess the problem	2	4	-	1
PC5. use the equipment data obtained from pressure gauges, filler gauges, callipers, condition monitoring devices and compare it with the standards to detect faults in the system	2	4	-	1
PC6. check the lubricant levels and refill as required, ensuring to use the lubricant of the correct code and grade	3	5	-	2
PC7. follow the Original Equipment Manufacturer (OEM)-recommended procedure and checklist for routine servicing	2	3	-	1
PC8. calibrate, align and adjust the settings, alignment, pressures, tension, speeds and levels of the engine, transmission system, load bearing arms and structure, electrical and electronic components, valves, ignition, fuel and emissions, brakes, lights, tires, steering and body fittings	3	4	-	1
PC9. identify wear and tear in components, such as oil and air filters, drive belts, braking system due to continuous usage and replace them with new ones	3	5	-	2

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. follow the appropriate measures to avoid damage to the vehicle and its components during repair and maintenance	1	3	-	2
PC11. carry out breakdown maintenance at the worksite or workshop, as required	3	5	-	2
PC12. follow appropriate safety measures while operating and maintaining the HEMM, and hydraulic machine	2	3	-	2
NOS Total	30	50	-	20

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	MIN/N1812
NOS Name	Carry out repair and maintenance of Heavy Earth Moving Machinery (HEMM)
Sector	Mining
Sub-Sector	Mining Operation
Occupation	Placer Mining
NSQF Level	4
Credits	3
Version	1.0
Last Reviewed Date	15/03/2024
Next Review Date	14/03/2027
NSQC Clearance Date	15/03/2024

Qualification Pack

MIN/N1813: Perform inventory and logistics management

Description

This unit is about performing inventory and logistics management.

Scope

The scope covers the following :

- Manage the inventory
- Inspect the work area and equipment

Elements and Performance Criteria

Manage the inventory

To be competent, the user/individual on the job must be able to:

- PC1.** check the types of inventories and obtain the list of stored items from the supervisor
- PC2.** identify the item categories as per the inventory classification schemes for inventory inspection
- PC3.** check the location and the level of items in each pallet
- PC4.** determine the number of items to be counted daily based on the total number of Stock Keeping Units (SKU) and the total working days in a month
- PC5.** follow the organizational procedure concerning materials handling and personnel safety during inventory management
- PC6.** follow the appropriate measures to mitigate and deal with fire, chemical and bio hazards in the warehouse

Inspect the work area and equipment

To be competent, the user/individual on the job must be able to:

- PC7.** perform inventory inspection wearing the appropriate Personal Protective Equipment (PPE), e.g. protective gear and helmet
- PC8.** ensure the availability of Material Handling Equipment (MHE) such as reach trucks and cages
- PC9.** inspect the MHE to ensure that they are in good working condition and ready-to-use
- PC10.** inspect the storage area for any spillage, breakage or any other safety issues
- PC11.** clean spills or breakages in the storage area
- PC12.** check the storage racks for overloading to ensure their stability

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** the types and functions of inventories
- KU2.** the identification of item categories as per the inventory classification schemes
- KU3.** the appropriate material handling procedure

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- KU4.** the appropriate personnel safety measures during inventory management
- KU5.** follow the appropriate measures to mitigate and deal with fire, chemical and bio hazards in the warehouse
- KU6.** the inventory inspection procedure
- KU7.** the inspection and use of appropriate Material Handling Equipment (MHE)
- KU8.** the importance of avoiding the overloading of storage racks
- KU9.** the importance of maintaining safe distance between the floor personnel and staying within the allotted areas

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** carry out appropriate documentation concerning work
- GS2.** read equipment manuals and process documents to understand the equipment and processes
- GS3.** listen attentively to understand the instructions or information being shared
- GS4.** communicate politely and professionally
- GS5.** plan and carry out different activities appropriately as per the applicable timelines and quality norms
- GS6.** follow the task lists and schedules
- GS7.** make appropriate decisions concerning the areas of work, following the organizational authority matrix
- GS8.** follow the given instructions and work on the identified areas of improvement
- GS9.** evaluate the possible solutions to work-related problems applying reasoning skills and select the best one
- GS10.** follow the organizational procedure during emergencies

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Manage the inventory</i>	16	26	-	11
PC1. check the types of inventories and obtain the list of stored items from the supervisor	3	5	-	2
PC2. identify the item categories as per the inventory classification schemes for inventory inspection	3	5	-	2
PC3. check the location and the level of items in each pallet	3	5	-	2
PC4. determine the number of items to be counted daily based on the total number of Stock Keeping Units (SKU) and the total working days in a month	3	5	-	2
PC5. follow the organizational procedure concerning materials handling and personnel safety during inventory management	2	3	-	2
PC6. follow the appropriate measures to mitigate and deal with fire, chemical and bio hazards in the warehouse	2	3	-	1
<i>Inspect the work area and equipment</i>	14	24	-	9
PC7. perform inventory inspection wearing the appropriate Personal Protective Equipment (PPE), e.g. protective gear and helmet	3	5	-	1
PC8. ensure the availability of Material Handling Equipment (MHE) such as reach trucks and cages	2	3	-	1
PC9. inspect the MHE to ensure that they are in good working condition and ready-to-use	2	4	-	2
PC10. inspect the storage area for any spillage, breakage or any other safety issues	2	4	-	2
PC11. clean spills or breakages in the storage area	2	3	-	1



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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC12. check the storage racks for overloading to ensure their stability	3	5	-	2
NOS Total	30	50	-	20

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	MIN/N1813
NOS Name	Perform inventory and logistics management
Sector	Mining
Sub-Sector	Mining Operation
Occupation	Placer Mining
NSQF Level	4
Credits	2
Version	1.0
Last Reviewed Date	15/03/2024
Next Review Date	14/03/2027
NSQC Clearance Date	15/03/2024

Qualification Pack

MIN/N1810: Follow the applicable health, safety and environmental guidelines (Placer Mining)

Description

This unit is about following the applicable health, safety and environmental guidelines.

Scope

The scope covers the following :

- Maintain personal health and safety
- Assist in hazard management
- Work on equipment and machinery
- Check the first aid box, firefighting and safety equipment
- Report incidents

Elements and Performance Criteria

Maintain personal health and safety

To be competent, the user/individual on the job must be able to:

- PC1.** follow the appropriate hand hygiene measures, i.e., washing hands with EDTA/Soda Ash (Ethylene Diamine Tetra Acetate/Sodium Carbonate), soap and water or alcohol-based hand sanitizers
- PC2.** ensure the plants are provided with a dilution ventilation system to avoid the undue accumulation of chemical vapours in working environments
- PC3.** clean and disinfect the regularly used objects, such as mobile phones, wireless communication systems keys, wallets and work passes
- PC4.** follow the Safety Management Plan (SMP) and Emergency Management Plan (EMP)
- PC5.** follow sanitation measures in the work area, e.g., avoiding eating, drinking, chewing and smoking
- PC6.** use the appropriate PPE, e.g. using an appropriate mask to cover the mouth and nose
- PC7.** identify six directional hazards at the workplace and take decisions accordingly
- PC8.** follow appropriate safety measures to avoid exposure to radiation, e.g. limiting stay in the radiation area and avoiding working with open wounds or exposed body parts in radiation areas
- PC9.** undertake the Take-5 (Personal Risk Assessment) before the commencement of any work
- PC10.** lift objects carefully maintaining the correct body posture to avoid injury
- PC11.** use self-rescue apparatus appropriately when required
- PC12.** work in the assigned work zone in the designated work area
- PC13.** follow the recommended safety measures while working at haul roads, heights, dumps, sump areas, stockyards, near moving parts, etc
- PC14.** wear Thermo luminescent Dosimeter (TLD) Badge for personal monitoring in rare earth extraction plant
- PC15.** use caution signs as applicable and obey safety signs, stickers, and tags

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PC16. identify the environmental impact of operations related to mining, mineral separation plant and rare earth extraction plant and take steps to reduce the impact

PC17. check for contamination of sensitive body parts, e.g. face, eyes etc., and seek immediate medical help

Assist in hazard management

To be competent, the user/individual on the job must be able to:

PC18. label and store chemicals appropriately

PC19. display the MSDS for the hazardous chemicals in storage and handling areas

PC20. educate the personnel at the workplace about the safety requirements and follow the applicable safety guidelines

PC21. check the body and clothing for contamination at the end of radioactive work, and decontaminate and take a shower at the decontamination centre

PC22. maintain regular and open communication with co-workers and the employer

Work on equipment and machinery

To be competent, the user/individual on the job must be able to:

PC23. check and ensure every instrument, apparatus and equipment is AERB approved

PC24. operate the appropriate tools, equipment and machinery with proper training and authorization

Check the first aid box, firefighting, and safety equipment

To be competent, the user/individual on the job must be able to:

PC25. check the availability of first aid kits and fire extinguishers, and ensure they are easily accessible

PC26. ensure the availability of emergency showers and eye washers in the chemical handling, unloading and pumping areas

PC27. monitor the work environment at appropriate intervals for hydrogen chloride and hydrocarbons using portable monitors

PC28. operate the appropriate type of fire extinguishers to control different types of fire at the worksite

Report incidents

To be competent, the user/individual on the job must be able to:

PC29. identify and report potential hazards/ unsafe conditions and practices

PC30. follow the appropriate emergency procedure during emergencies, such as power failures, cyclones, etc.

PC31. report the emergencies and any safety incidents as per the organizational procedure

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. different types of chemical plants and their operations

KU2. the importance of time-keeping, discipline and punctuality

KU3. various chemical processes carried out in a rare earth extraction plant

KU4. the impact of violating applicable safety procedures

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- KU5.** the relevant quality norms
- KU6.** different types of individual rare earth chemical processes and the use of associated equipment
- KU7.** different types of effluent treatment processes and the use of associated equipment
- KU8.** the use of different water purifying and waste treatment agents
- KU9.** the hazards and the appropriate safety measures in acid/ alkali handling activities and usage of relevant PPE
- KU10.** hazards and safety aspects involved in acid/ alkali handling activities and usage of relevant PPE
- KU11.** the internal safety organization and the role of the safety committee, workforce inspector and AERB, etc
- KU12.** the importance of sensitization towards different genders and persons with disabilities (PWD)
- KU13.** the Importance of following infection control policies, '5-S' practices, and waste management as prescribed by regulatory authorities like the Atomic Energy Regulatory Board (AERB), United Nations Development Group (UNDG), etc.
- KU14.** the importance of water/material/energy conservation and management
- KU15.** the importance of Social Management Plan (SMP) and Environment Management Plan (EMP) prepared by the organization, as per DGMS guideline
- KU16.** the importance of wearing own TLD or pocket dosimeter (to be obtained from the Health Physics Unit) in the active area
- KU17.** the importance of staying alert and attentive to the work surroundings
- KU18.** the importance of staying in the assigned work zone in the designated work area
- KU19.** the benefits of making the safety rules part of new employee orientation
- KU20.** the importance of contacting the supervisor or manager for instruction, guidance, or training when required

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** complete appropriate documentation concerning work
- GS2.** make appropriate decisions concerning the area of work with effective reasoning
- GS3.** read and explain manuals, health, and safety instructions, memos, etc
- GS4.** plan and organize work effectively
- GS5.** detect problems at work and use reasoning skills to resolve them promptly and effectively
- GS6.** complete the assigned tasks timely

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Maintain personal health and safety</i>	16	24	-	11
PC1. follow the appropriate hand hygiene measures, i.e., washing hands with EDTA/Soda Ash (Ethylene Diamine Tetra Acetate/Sodium Carbonate), soap and water or alcohol-based hand sanitizers	1	2	-	1
PC2. ensure the plants are provided with a dilution ventilation system to avoid the undue accumulation of chemical vapours in working environments	1	1	-	1
PC3. clean and disinfect the regularly used objects, such as mobile phones, wireless communication systems keys, wallets and work passes	1	1	-	-
PC4. follow the Safety Management Plan (SMP) and Emergency Management Plan (EMP)	1	2	-	-
PC5. follow sanitation measures in the work area, e.g., avoiding eating, drinking, chewing and smoking	1	2	-	-
PC6. use the appropriate PPE, e.g. using an appropriate mask to cover the mouth and nose	1	2	-	-
PC7. identify six directional hazards at the workplace and take decisions accordingly	1	2	-	1
PC8. follow appropriate safety measures to avoid exposure to radiation, e.g. limiting stay in the radiation area and avoiding working with open wounds or exposed body parts in radiation areas	1	2	-	1
PC9. undertake the Take-5 (Personal Risk Assessment) before the commencement of any work	1	2	-	1
PC10. lift objects carefully maintaining the correct body posture to avoid injury	-	1	-	-
PC11. use self-rescue apparatus appropriately when required	1	1	-	1

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC12. work in the assigned work zone in the designated work area	1	1	-	1
PC13. follow the recommended safety measures while working at haul roads, heights, dumps, sump areas, stockyards, near moving parts, etc	1	1	-	1
PC14. wear Thermo luminescent Dosimeter (TLD) Badge for personal monitoring in rare earth extraction plant	1	1	-	1
PC15. use caution signs as applicable and obey safety signs, stickers, and tags	1	1	-	-
PC16. identify the environmental impact of operations related to mining, mineral separation plant and rare earth extraction plant and take steps to reduce the impact	1	1	-	1
PC17. check for contamination of sensitive body parts, e.g. face, eyes etc., and seek immediate medical help	1	1	-	1
<i>Assist in hazard management</i>	5	8	-	4
PC18. label and store chemicals appropriately	1	1	-	1
PC19. display the MSDS for the hazardous chemicals in storage and handling areas	1	2	-	1
PC20. educate the personnel at the workplace about the safety requirements and follow the applicable safety guidelines	1	1	-	-
PC21. check the body and clothing for contamination at the end of radioactive work, and decontaminate and take a shower at the decontamination centre	1	2	-	1
PC22. maintain regular and open communication with co-workers and the employer	1	2	-	1
<i>Work on equipment and machinery</i>	2	4	-	1
PC23. check and ensure every instrument, apparatus and equipment is AERB approved	1	2	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC24. operate the appropriate tools, equipment and machinery with proper training and authorization	1	2	-	1
<i>Check the first aid box, firefighting, and safety equipment</i>	4	10	-	2
PC25. check the availability of first aid kits and fire extinguishers, and ensure they are easily accessible	1	3	-	-
PC26. ensure the availability of emergency showers and eye washers in the chemical handling, unloading and pumping areas	1	2	-	-
PC27. monitor the work environment at appropriate intervals for hydrogen chloride and hydrocarbons using portable monitors	1	2	-	1
PC28. operate the appropriate type of fire extinguishers to control different types of fire at the worksite	1	3	-	1
<i>Report incidents</i>	3	4	-	2
PC29. identify and report potential hazards/ unsafe conditions and practices	1	2	-	-
PC30. follow the appropriate emergency procedure during emergencies, such as power failures, cyclones, etc.	1	1	-	1
PC31. report the emergencies and any safety incidents as per the organizational procedure	1	1	-	1
NOS Total	30	50	-	20

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	MIN/N1810
NOS Name	Follow the applicable health, safety and environmental guidelines (Placer Mining)
Sector	Mining
Sub-Sector	Mining Operation
Occupation	Placer Mining
NSQF Level	4
Credits	1
Version	1.0
Last Reviewed Date	15/03/2024
Next Review Date	14/03/2027
NSQC Clearance Date	15/03/2024

Qualification Pack

MIN/N1811: Work collaboratively and effectively in a team

Description

This unit is about the importance of teamwork in the workplace.

Scope

The scope covers the following :

- Work effectively with co-workers
- Communicate effectively with co-workers
- Practice inclusiveness at work

Elements and Performance Criteria

Work effectively with co-workers

To be competent, the user/individual on the job must be able to:

- PC1.** identify own direct and indirect responsibilities, e.g. duties concerning team members
- PC2.** undertake appropriate activities to promote cooperation and good relationships with co-workers
- PC3.** mentor and assist subordinates in the execution of their work responsibilities
- PC4.** identify possible disruptions to work through coordination with the relevant stakeholders and take appropriate preventive measures
- PC5.** follow the recommended practices to avoid and resolve conflicts at work
- PC6.** interact effectively and cooperate with co-workers
- PC7.** manage workplace conflicts, ensuring no impact on work
- PC8.** identify and take responsibility for own mistakes
- PC9.** follow the organizational policies and procedures
- PC10.** follow the appropriate productivity enhancement measures, e.g. Quality circle, 5s, TPM, Six Sigma, JIT, etc.
- PC11.** exchange feedback with peers and other team members to improve the quality of work

Communicate effectively with co-workers

To be competent, the user/individual on the job must be able to:

- PC12.** follow the organizational policy for the efficient and timely dissemination of information to the authorized personnel
- PC13.** communicate clearly and politely to ensure effective communication with co-workers
- PC14.** practice active listening to understand the conveyed information and instructions
- PC15.** practice openness in work conversations, giving undivided attention to enhance the quality of workplace communication
- PC16.** maintain the professional standard of communication, e.g. avoiding the use of offensive language at work

Practice inclusiveness at work

To be competent, the user/individual on the job must be able to:

Qualification Pack

- PC17.** empathize with Persons with Disabilities (PwD)
- PC18.** practice gender equality at work
- PC19.** maintain respectful behaviour and etiquette at work
- PC20.** listen to and acknowledge the feelings, concerns, opinions, and ideas of others

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** the importance of teamwork and the consequence of poor teamwork on project outcomes, timelines, safety, etc
- KU2.** the duties and rights of workers
- KU3.** the importance of creating a healthy and cooperative work environment
- KU4.** the importance of proper and effective communication and the adverse effects of poor communication on quality, timeliness, safety, etc
- KU5.** different modes of communication used at work
- KU6.** the importance and need of supporting co-workers facing problems for the smooth functioning of work
- KU7.** the fundamental concept of gender equality
- KU8.** how to recognize and be sensitive to issues of disability, culture and gender
- KU9.** how to show empathy to PwD appropriately
- KU10.** legislation, policies, and procedures relating to gender sensitivity and cultural diversity including their impact on the area of operation
- KU11.** the appropriate behaviour to be practised with coworkers
- KU12.** the appropriate communication skills and leadership qualities
- KU13.** how to be open to listening to and understanding the other person's point of view

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** complete appropriate documentation concerning work
- GS2.** make appropriate decisions concerning the area of work with effective reasoning
- GS3.** read and explain manuals, health, and safety instructions, memos, etc
- GS4.** plan and organize work effectively
- GS5.** detect problems at work and use reasoning skills to resolve them promptly and effectively
- GS6.** complete the assigned tasks timely

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Work effectively with co-workers</i>	15	30	-	10
PC1. identify own direct and indirect responsibilities, e.g. duties concerning team members	3	5	-	3
PC2. undertake appropriate activities to promote cooperation and good relationships with co-workers	1	2	-	1
PC3. mentor and assist subordinates in the execution of their work responsibilities	1	2	-	1
PC4. identify possible disruptions to work through coordination with the relevant stakeholders and take appropriate preventive measures	3	4	-	2
PC5. follow the recommended practices to avoid and resolve conflicts at work	1	2	-	1
PC6. interact effectively and cooperate with co-workers	1	1	-	-
PC7. manage workplace conflicts, ensuring no impact on work	1	1	-	-
PC8. identify and take responsibility for own mistakes	1	3	-	1
PC9. follow the organizational policies and procedures	1	3	-	-
PC10. follow the appropriate productivity enhancement measures, e.g. Quality circle, 5s, TPM, Six Sigma, JIT, etc.	1	4	-	1
PC11. exchange feedback with peers and other team members to improve the quality of work	1	3	-	-
<i>Communicate effectively with co-workers</i>	9	10	-	5
PC12. follow the organizational policy for the efficient and timely dissemination of information to the authorized personnel	2	3	-	2

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC13. communicate clearly and politely to ensure effective communication with co-workers	1	2	-	-
PC14. practice active listening to understand the conveyed information and instructions	2	1	-	1
PC15. practice openness in work conversations, giving undivided attention to enhance the quality of workplace communication	2	2	-	1
PC16. maintain the professional standard of communication, e.g. avoiding the use of offensive language at work	2	2	-	1
<i>Practice inclusiveness at work</i>	6	10	-	5
PC17. empathize with Persons with Disabilities (PwD)	2	2	-	2
PC18. practice gender equality at work	2	2	-	2
PC19. maintain respectful behaviour and etiquette at work	1	3	-	1
PC20. listen to and acknowledge the feelings, concerns, opinions, and ideas of others	1	3	-	-
NOS Total	30	50	-	20

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	MIN/N1811
NOS Name	Work collaboratively and effectively in a team
Sector	Mining
Sub-Sector	Mining Operation
Occupation	Placer Mining
NSQF Level	4
Credits	1
Version	1.0
Last Reviewed Date	15/03/2024
Next Review Date	14/03/2027
NSQC Clearance Date	15/03/2024

Qualification Pack

DGT/VSQ/N0102: Employability Skills (60 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Career Development & Goal Setting
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

- PC1.** identify employability skills required for jobs in various industries
- PC2.** identify and explore learning and employability portals

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

- PC3.** recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.
- PC4.** follow environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

- PC5.** recognize the significance of 21st Century Skills for employment
- PC6.** practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life

Basic English Skills

To be competent, the user/individual on the job must be able to:

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- PC7.** use basic English for everyday conversation in different contexts, in person and over the telephone
- PC8.** read and understand routine information, notes, instructions, mails, letters etc. written in English
- PC9.** write short messages, notes, letters, e-mails etc. in English

Career Development & Goal Setting

To be competent, the user/individual on the job must be able to:

- PC10.** understand the difference between job and career
- PC11.** prepare a career development plan with short- and long-term goals, based on aptitude

Communication Skills

To be competent, the user/individual on the job must be able to:

- PC12.** follow verbal and non-verbal communication etiquette and active listening techniques in various settings
- PC13.** work collaboratively with others in a team

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

- PC14.** communicate and behave appropriately with all genders and PwD
- PC15.** escalate any issues related to sexual harassment at workplace according to POSH Act

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

- PC16.** select financial institutions, products and services as per requirement
- PC17.** carry out offline and online financial transactions, safely and securely
- PC18.** identify common components of salary and compute income, expenses, taxes, investments etc
- PC19.** identify relevant rights and laws and use legal aids to fight against legal exploitation

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

- PC20.** operate digital devices and carry out basic internet operations securely and safely
- PC21.** use e- mail and social media platforms and virtual collaboration tools to work effectively
- PC22.** use basic features of word processor, spreadsheets, and presentations

Entrepreneurship

To be competent, the user/individual on the job must be able to:

- PC23.** identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research
- PC24.** develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion
- PC25.** identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity

Customer Service

To be competent, the user/individual on the job must be able to:

- PC26.** identify different types of customers
- PC27.** identify and respond to customer requests and needs in a professional manner.

Qualification Pack

PC28. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC29. create a professional Curriculum vitae (Résumé)

PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively

PC31. apply to identified job openings using offline /online methods as per requirement

PC32. answer questions politely, with clarity and confidence, during recruitment and selection

PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills and different learning and employability related portals

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use English language for effective verbal (face to face and telephonic) and written communication in formal and informal set up

KU6. importance of career development and setting long- and short-term goals

KU7. about effective communication

KU8. POSH Act

KU9. Gender sensitivity and inclusivity

KU10. different types of financial institutes, products, and services

KU11. how to compute income and expenditure

KU12. importance of maintaining safety and security in offline and online financial transactions

KU13. different legal rights and laws

KU14. different types of digital devices and the procedure to operate them safely and securely

KU15. how to create and operate an e- mail account and use applications such as word processors, spreadsheets etc.

KU16. how to identify business opportunities

KU17. types and needs of customers

KU18. how to apply for a job and prepare for an interview

KU19. apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. read and write different types of documents/instructions/correspondence

GS2. communicate effectively using appropriate language in formal and informal settings



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- GS3.** behave politely and appropriately with all
- GS4.** how to work in a virtual mode
- GS5.** perform calculations efficiently
- GS6.** solve problems effectively
- GS7.** pay attention to details
- GS8.** manage time efficiently
- GS9.** maintain hygiene and sanitization to avoid infection

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. identify employability skills required for jobs in various industries	-	-	-	-
PC2. identify and explore learning and employability portals	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC3. recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.	-	-	-	-
PC4. follow environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	2	4	-	-
PC5. recognize the significance of 21st Century Skills for employment	-	-	-	-
PC6. practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life	-	-	-	-
<i>Basic English Skills</i>	2	3	-	-
PC7. use basic English for everyday conversation in different contexts, in person and over the telephone	-	-	-	-
PC8. read and understand routine information, notes, instructions, mails, letters etc. written in English	-	-	-	-
PC9. write short messages, notes, letters, e-mails etc. in English	-	-	-	-
<i>Career Development & Goal Setting</i>	1	2	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. understand the difference between job and career	-	-	-	-
PC11. prepare a career development plan with short- and long-term goals, based on aptitude	-	-	-	-
<i>Communication Skills</i>	2	2	-	-
PC12. follow verbal and non-verbal communication etiquette and active listening techniques in various settings	-	-	-	-
PC13. work collaboratively with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	2	-	-
PC14. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC15. escalate any issues related to sexual harassment at workplace according to POSH Act	-	-	-	-
<i>Financial and Legal Literacy</i>	2	3	-	-
PC16. select financial institutions, products and services as per requirement	-	-	-	-
PC17. carry out offline and online financial transactions, safely and securely	-	-	-	-
PC18. identify common components of salary and compute income, expenses, taxes, investments etc	-	-	-	-
PC19. identify relevant rights and laws and use legal aids to fight against legal exploitation	-	-	-	-
<i>Essential Digital Skills</i>	3	4	-	-
PC20. operate digital devices and carry out basic internet operations securely and safely	-	-	-	-
PC21. use e- mail and social media platforms and virtual collaboration tools to work effectively	-	-	-	-
PC22. use basic features of word processor, spreadsheets, and presentations	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Entrepreneurship</i>	2	3	-	-
PC23. identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research	-	-	-	-
PC24. develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion	-	-	-	-
PC25. identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity	-	-	-	-
<i>Customer Service</i>	1	2	-	-
PC26. identify different types of customers	-	-	-	-
PC27. identify and respond to customer requests and needs in a professional manner.	-	-	-	-
PC28. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	2	3	-	-
PC29. create a professional Curriculum vitae (Résumé)	-	-	-	-
PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively	-	-	-	-
PC31. apply to identified job openings using offline /online methods as per requirement	-	-	-	-
PC32. answer questions politely, with clarity and confidence, during recruitment and selection	-	-	-	-
PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements	-	-	-	-
NOS Total	20	30	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0102
NOS Name	Employability Skills (60 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	4
Credits	2
Version	1.0
Last Reviewed Date	12/03/2026
Next Review Date	12/03/2029
NSQC Clearance Date	12/03/2026

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

1. Criteria for assessment for each Qualification Pack will be created by the Sector Skill Council. Each Element/ Performance Criteria (PC) will be assigned marks proportional to its importance in NOS. SSC will also lay down proportion of marks for Theory and Skills Practical for each Element/ PC.
2. The assessment for the theory part will be based on knowledge bank of questions created by the SSC.
3. Assessment will be conducted for all compulsory NOS, and where applicable, on the selected elective/option NOS/set of NOS.
4. Individual assessment agencies will create unique question papers for theory part for each candidate at each examination/training center (as per assessment criteria below).
5. Individual assessment agencies will create unique evaluations for skill practical for every student at each examination/ training center based on these criteria.
6. To pass the Qualification Pack assessment, every trainee should score the Recommended Pass % aggregate for the QP.
7. In case of unsuccessful completion, the trainee may seek reassessment on the Qualification Pack.

Qualification Pack

Minimum Aggregate Passing % at QP Level : 70

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
MIN/N4107.Carry out the dredging operation	30	50	0	20	100	25
MIN/N4108.Carry out the upgradation unit operations	30	50	0	20	100	25
MIN/N1812.Carry out repair and maintenance of Heavy Earth Moving Machinery (HEMM)	30	50	0	20	100	20
MIN/N1813.Perform inventory and logistics management	30	50	0	20	100	15
MIN/N1810.Follow the applicable health, safety and environmental guidelines (Placer Mining)	30	50	0	20	100	5
MIN/N1811.Work collaboratively and effectively in a team	30	50	0	20	100	5
DGT/VSQ/N0102.Employability Skills (60 Hours)	20	30	-	-	50	5
Total	200	330	-	120	650	100

Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training

Qualification Pack

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

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Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.